VERA — Vision-Enabled Reasoning Agent — System Architecture

v1.92.1 — Web UI: Vision title in the camera top bar

Root-owned deployment controller (separate from VERA)
vera update + authenticated System GUI /run/vera-updater/control.sock peer-checked fixed operations
vera-updater.timer vera-updater-check.service local-time installation window
local Unix socket
vera-updater.service /opt/vera-updater/src/updater stage / deploy / verify / recover
ZeroMQ IPC
verified full-duty kernel PWM before thermal restart
manifest + verified assets
atomic journal
deployment.readiness RPC loaded version/revision/PID fresh telemetry + service/hardware health
NF-A6x25 PWM Fan PWM ← GPIO13 (kernel HW PWM, 25 kHz) TACH → GPIO6 (2 ppr)
GitHub stable releases mazum01/desktop-assistant HTTPS, exact tag/commit
/var/lib/vera-updater protected backups + transaction state post-reboot continuation
tach
Optional Matter processes (systemd user services, Node.js 24)
vera-matter-controller OHF matter-server local dashboard/WS 127.0.0.1:5580
Google Home Matter controller
Matter / IPv6 LAN
independent Matter fabric
Matter / IPv6 LAN
Tapo P316M / Matter Wi-Fi devices native multi-admin sharing
vera-matter-bridge src/matter/bridge.mjs VERA Audio Mute, UDP 5541
~/.local/share/vera-matter separate controller + bridge fabrics private persistent credentials
Hardware (Raspberry Pi 5)
TMP102 (VERA, I²C 0x48)
Pi Cam Module 3 Wide (CSI-0)
TMP117 (VERA3, I²C 0x48)
Hailo AI Accelerator (Hailo-8 / Hailo-10H, PCIe)
Pan servo + optional tilt PCA9685 configurable channels 180deg direct-drive / legacy DS3218 configured safe travel limits
reSpeaker XVF3800 (USB mic array)
reSpeaker XVF3800 output → attached speaker
Waveshare ESP32-S3-Touch-LCD-1.47 172×320 SPI TFT (touch disabled) (BLE mouth/status display + BLE OTA) C6 touch + legacy profiles supported
Display microSD (S3) /videos/*.mjpg or *.mjpeg baseline JPEG streams, no audio
Optional QMI8658 (identity-probed) not specified on S3 hardware roll/pitch/relative yaw, no compass
CSI
I²C
PCIe control (temp only)
PCIe
I²C/PWM
USB
I²C
PWM
SPI + GPIO
SD_MMC 1-bit CLK16/CMD15/D0=17
Drivers (src/)
vision/camera.py (picamera2)
thermal/hailo_temp.py (Hailo die temp via HailoRT control, monitor only)
thermal/tmp117.py
audio/version_announcer.py
perception/hailo_inference.py (VStreams + InferModel API)
perception/hailo_models.py config/hailo/models.yaml probe / select / download HEFs python3 -m ...hailo_models --install
motion/servo_controller.py
motion/visual_servo.py (optional controller) stable target lock + damped visual feedback identified camera/servo Jacobian config/tracking_calibration.json (local)
core/body.py config/body.yaml (machine-local identity/tuning) config/body_profiles.yaml (hardware/controller) body-tagged runtime servo geometry
motion/head_tracker.py (original body) legacy Kalman + minimum-jerk pan tracking new-body idle motion controller
motion/gaze_tracker.py (new body default) world-bearing Kalman + minimum-jerk pan/tilt smooth pursuit · fixation hold · saccadic suppression
audio/input.py
TMP117 ID
matched SCRFD HEF
thermal/fan_tach.py (lgpio edge cb)
thermal/fan.py (sysfs HW PWM + lgpio fallback)
thermal/temperature_sensor.py (probes TMP117 / TMP102)
audio/tts.py (espeak-ng)
perception/object_detector.py (YOLOv11l on-chip NMS; YOLO26m+ONNX H8 fallback)
perception/face_detector.py (SCRFD decode + Haar fallback)
thermal/thermal_manager.py
audio/output.py
renders to
External processes
scripts/desktop-assistant CLI: ping / status / pan / say / watch / ... display video list/play/stop/status waits for correlated device reply
~/.local/share/desktop-assistant/ telemetry.db (SQLite ring) faces.db (identity registry) thumbs/ (64×64 JPEG face crops)
3rd-party subscribers (dashboards, future ML pipelines)
Web Browser http://<pi>:8080
firmware/upload_esp32_display_ble.sh + nimbleota.py uploader (BLE firmware push, no USB)
ZMQ REQ/SUB
writes telemetry.db
ZMQ SUB
loopback GET/PUT /api/audio/mute
HTTP/WS :8080
latest_frame()
latest_observation()
latest_frame()
writes faces.db
latest_frame()
latest_frame()
matched ArcFace HEF
ZMQ PUB→SUB (cross-process)
vision.mono_depth_map
IPCClient RPC strict music.get_muted / set_muted RPC (music/podcast reads + actions)
ZMQ PUB→SUB (cross-process)
wiring, range, limits, tilt capability
controller and mount direction
USB
matched depth HEF
BLE GATT (OTA)
VisionService / RawCameraService → vision.frame_ready / vision.frame2_ready ← vision.capture_still RPC vision.get_focus / set_focus RPC camera2.get_focus / set_focus Web: GET|PUT /api/settings/camera{,2}/focus per-device persisted AF / manual diopters + latest_frame()
TelemetryService ← thermal.* / motion.position / audio.level → telemetry.flush SQLite ring buffer
IPCBridge (core) PUB ipc:///tmp/desktop-assistant.pub REP ipc:///tmp/desktop-assistant.rep SUB ← thermal.pub (upstream) cmds: ping / topics / last / publish / status
AVService ← av.say / beep / utterance / announce_version → av.spoke / av.speaking_started / version_announced
ObjectService ← vision.frame_ready / object.set_enabled ← perception.faces (selective label association) → perception.objects (held overlays + fresh persons) → object.enabled_changed (startup + changes) 2fps · overlay max_objects cap · BGR→RGB fix
WebService (FastAPI) GET / — dashboard SPA (hash routes, ES modules) GET /stream — MJPEG 640×360 feed WS /ws — live JSON status GET /api/faces — list (has_thumb flag) GET /api/skills — registered skills GET /api/faces/{id}/thumb — face JPEG PUT /api/faces/{id} — rename DEL /api/faces/{id} — delete one DEL /api/faces — delete all GET|PUT /api/settings/quiet-hours GET|PUT /api/settings/units (display_units.json) GET|PUT /api/settings/audio GET|PUT /api/settings/voice GET|PUT /api/settings/servo/channels PUT /api/settings/servo/tilt-limits GET|PUT /api/room GET|PUT /api/music/* — proxied via IPCClient GET|POST /api/podcasts/* — proxied via IPCClient POST /api/say|pan|tilt|version|utterance port 8080
ThermalService — thermal.temp — thermal.fan — thermal.rpm — thermal.critical — thermal.error — thermal.hailo
PerceptionService ← vision.frame_ready → perception.faces / perception.error + FaceEmbedder (ArcFace) + FaceRegistry
MotionService shared PCA9685, independent pan / tilt 50Hz bounded quintic profiles ← motion.pan_to / tilt_to / center / relax / stop ← motion.calibration_lock (expiring lease) ← motion.set_channels / set_limits / set_tilt_limits → motion.position / moved / settings / error → motion.channels_changed / limits_changed / tilt_limits_changed RPC motion.get_status / motion.set_channels
TrackingService (body-selected controller) original: legacy Kalman/minimum-jerk pan only new: gaze (Kalman bearing) pan/tilt; visual_servo optional ← perception.faces / perception.objects (fresh persons) ← object.enabled_changed (person-seek gate) ← motion.position / settings / limits_changed Enabled + ready servos required Recognized / largest face; 1s loss handoff Largest person + 15% stability margin → motion.pan_to / tilt_to (new body only) ← tracking.start_autotune / cancel_autotune (visual_servo) → motion.calibration_lock / av.say ← av.spoke (announce before probes) → tracking.debug / autotune_progress / autotune_done Tilt: travel limits and center +/-20deg independent random idle pan · speaking nod
AudioCaptureService → audio.level / chunk + latest_chunk()
latest_chunk() + chunk_index()
VoiceCommandService ← audio stream/VAD → voice.state / wake / transcript / intent → av.utterance (routed transcript) wake-window + dialog context
IPCBridge (media) PUB ipc:///tmp/desktop-assistant-media.pub REP ipc:///tmp/desktop-assistant-media.rep SUB ← core.pub (upstream) RPCs: music.get_state/set_volume/... podcast.search/subscribe/play/...
systemd: desktop-assistant-core.service
RoomService ← room.set → room.updated / av.say persists room name · visual scene baseline histogram · sustained divergence detection → TTS prompt
MonoDepthService scdepthv3 on Hailo-8 / Hailo-10H → vision.mono_depth_map relative 320×256 depth, 3 Hz
Arduino firmware firmware/vera_display (ESP32-S3 process boundary; C6 profiles retained) sd_video + JPEGDEC / orientation fusion main-loop queue + reserved angle footer NimBLEOta service (BLE OTA, 0x8018)
BLE GATT
BLE notifications
replies / playback events / IMU status
DisplayService ← system.startup_status / service.started / *.error ← display.spectrum / av.speaking_started / av.spoke ← display.video (list/play/stop/status) → display.status / display.error / display.reply → display.reply.<request_id> / display.device_event → display.spectrum_latency (audio→BLE→ack timing) BLE command writes (MTU; eq frames write-without-response)
PlaybackSpectrumService ← music.state_changed / podcast.playback pw-record on DA EQ sink monitor (pre sync-delay) + FFT → display.spectrum ← display.spectrum_latency → AvSyncCalibrator pw-cli set-param EQ sync_delay (A/V sync)
FaceService ← perception.faces / face.meet → av.say (greetings) → face.identified cooldown-based re-greet
ClockService + ClockAnnouncer → av.say (hourly/half-hourly) respects QuietHours
SkillsService ← av.utterance / thermal.temp / thermal.hailo → av.say / vision.describe / av.tell_joke → face.meet / motion.pan_to / music.* → object.set_enabled / music.set_volume → tracking.set_face_tracking SkillRegistry — first-match dispatch 17 skills: help · system_status · volume quiet_hours · weather · reminder smart_home · news + 9 others ConfigField framework: enabled / config_schema
NotificationService ← thermal.temp / thermal.hailo / perception.faces ← av.speaking_started / av.spoke → av.say (thermal, Hailo, absence alerts) rate-limited · quiet-hours aware
systemd: desktop-assistant-thermal.service
IPCBridge (thermal) PUB ipc:///tmp/desktop-assistant-thermal.pub REP ipc:///tmp/desktop-assistant-thermal.rep
MusicService ← music.play / stop / pause / next ← music.thumbs_up / thumbs_down / set_volume Pandora / audio playback
PodcastService Apple Podcasts search + RSS subscriptions ← podcast.play/pause/resume/stop/refresh → podcast.playback / subscriptions_updated
extends
config R/W
suppresses greetings
suppresses announcements
suppresses alerts
Core (src/core/)
core MessageBus (in-process pub/sub)
version.py (reads /VERSION)
thermal MessageBus (in-process pub/sub)
Service base (start/tick/stop)
QuietHours (enabled / start / end loaded from config/quiet_hours.yaml)
media MessageBus (in-process pub/sub)